

CS445 - NATURAL LANGUAGE PROCESSING

Spring 2026

Instructor:	Dilara Keküllüoğlu	Email:	dilara.kekulluoglu@sabanciuniv.edu
Time:	M 12:40 - 14:30, W 16:40 - 17:30	Place:	FENS L045

TAs: TBA

Office Hours:

Dilara Keküllüoğlu Wednesday 10:00-11:00 FENS 1089

You can also come visit the instructor in their office by appointment through email. Please add CS445 in the subject in your emails so we do not miss them.

Main Reference: Daniel Jurafsky and James H. Martin, *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition* (3rd edition online). <https://web.stanford.edu/~jurafsky/slp3/>

Optional Reference: Steven Bird, Ewan Klein, and Edward Loper, *Natural Language Processing with Python*. <https://www.nltk.org/book/>

Contents: This course studies the theory, design, and implementation of natural language processing systems. Topics include text processing, regular expressions, statistical properties of text, edit distance, language modeling, text classification, sequence modeling, topic modeling, computational morphology, neural networks for NLP, chatbots, and transfer learning for NLP.

Learning Objectives: This course aims to give foundational knowledge of the Natural Language Processing field. A student who successfully fulfills the course requirements will be able to:

1. **describe** the statistical properties of text in natural language.
2. **implement** programs that can process textual data and extract valuable information from it.
3. **apply** well-known language processing techniques to text.
4. **explain** the significance and principles of language modeling.
5. **develop** machine learning models to classify documents, sub-documents, or terms.
6. **assess** the quality of natural language processing models applied to text.

Prerequisites: CS 204 and (CS 210 or DSA 210)

Course Outline:

- Week 1: Introduction to Natural Language Processing
- Week 2: Regular Expressions and Text Normalization
- Week 3: Text Analysis
- Week 4: Language Models
- Week 5: Text Classifications and Neural Networks
- Week 6: Semantics and Embeddings
- Week 7: Midterm I (Tentative)
- Week 8: Morphology, POS Tagging, and Named Entities
- Week 9: Hidden Markov Models
- Week 10: Context-Free Grammars and Parsing
- Week 11: Large Language Models
- Week 12: Conversational Agents
- Week 13: Midterm II (Tentative)
- Week 14: NLP as a tool in Current Research

Grading Policy: Coding Assignments (15%), Midterms(60%), Project (25%).

Important Dates: These dates are tentative and subject to change. We will notify you of the exact dates for the midterm exams.

Midterm I	Week 7
Midterm II	Week 13
Project Deadline	Week 14

Class Policy:

- Regular attendance is essential and expected. Attendance will be taken but will not count towards your final grade.
- You have three points you can use for late submissions. Every day you are late for a coding assignment will remove a point from your balance. You can use these three points as you wish for the coding assignments. If you do not have any points left, you will receive zero marks for missed deadlines.
- Late submissions are not accepted for the project deadlines.
- Midterms will include all topics covered in class up to the exam date.
- We will have only one makeup exam, which may count only for one of the midterm exams. Missing both of the midterms will result in an automatic fail.

Grading:

- Written exams, coding assignments, and the projects will be counted for the last letter grade.

- Letter grades will be assigned based on the curve, considering the distribution of the grades in the class. If the class average is too high, we can apply catalogue grades.
- Any one of the following leads to failing the course automatically: (1) overall total grade below 40 (over 100), (2) any exam grade below 30 (over 100), (3) missing both of the midterm exams. However, not having an automatic fail does not guarantee you will pass the course (e.g., you may still fail with a total grade over 40, depending on the course grade distribution).

Extra Credits from Research Participation - up to 3 points

Students can convert their SONA research points into bonus points for this course. You can collect SONA points by participating in research conducted by Sabanci University researchers. You will be able to sign up for experiments and earn your Research Points through the online Sona at <https://sabanciuniv.sona-systems.com/> The system details can be found in Research Points - GuideForStudents Spring 2026.pdf on SUCourse.

You may convert up to 6 SONA points into 3 course points. If you use SONA credits for a bonus, you cannot participate in other bonus assignments we might have in the course (there may be at most one with potential for three points). However, we may also have no additional bonus assignments.

Generative AI Policy

Students may use AI tools under specific conditions. For assignments and group projects, you are not allowed to write the code from scratch with an AI tool. You may use AI tools to give you suggestions for functionalities you want and to debug your code, like you would use Google Search (e.g., what library to use for spellchecking, how to fix XXX error). However, you should not generate the entirety of the code with an AI tool (e.g., write an intent detection system for me). You are responsible for your own submissions. Please double-check and test the suggestions provided by the AI tool to ensure they are working correctly. For the project reports, AI tools may only be used for editing purposes (e.g., grammar and clarity). Generating content, such as writing sections of the report, structuring arguments, or producing analysis with AI, is not permitted. No distinction is made between different AI platforms; all tools are treated equally under this policy. You are allowed to upload course materials to AI tools for studying, but do not upload the course materials online publicly.

This approach is intended to support students' learning and writing skills while ensuring that the core analytical and conceptual work in the term project reflects their own understanding. For any questions about the AI policy, students should contact the course instructor directly via email or during office hours. In addition, students should be informed of the "Guidelines on the Use of Generative AI", which outlines Sabanci University's general approach to the use of AI in education.

Academic Honesty: All students must follow the university guidelines of academic integrity. <https://www.sabanciuniv.edu/en/academic-integrity-statement>

Flexibility: The contents of the syllabus are subject to change. The students will be notified of any updates.